

HOMEWORK 1

1.Sum of two matrices

CODE:

```
#include <stdio.h>

int main(){

    int mat1[10][10], mat2[10][10], rows, cols, i, j;


    printf("Enter size of row:");

    scanf("%d",&rows);


    printf("Enter size of column:");

    scanf("%d",&cols);


    printf("Enter elements in matrix 1:\n");

    for(i=0; i<rows; i++){

        for(j=0; j<cols; j++){

            scanf("%d",&mat1[i][j]);

        }

    }


    printf("Enter elements in matrix 2:\n");

    for(i=0; i<rows; i++){

        for(j=0; j<cols; j++){

            scanf("%d",&mat2[i][j]);
```

```
    }  
}
```

```
printf("\n");  
printf("Matrix 1:\n");  
for(i=0; i<rows; i++){  
    for(j=0; j<cols; j++){  
        printf("%d ",mat1[i][j]);  
    }  
    printf("\n");  
}
```

```
printf("\n");  
printf("Matrix 2:\n");  
for(i=0; i<rows; i++){  
    for(j=0; j<cols; j++){  
        printf("%d ",mat2[i][j]);  
    }  
    printf("\n");  
}
```

```
printf("\n");  
printf("Sum of two matrices:\n");  
for(i=0; i<rows; i++){  
    for(j=0; j<cols; j++){
```

```
        printf("%d ",mat1[i][j] + mat2[i][j]);  
    }  
    printf("\n");  
}  
  
return 0;  
}
```

OUTPUT:

Enter size of row:2

Enter size of column:2

Enter elements in matrix 1:

1

2

3

4

Enter elements in matrix 2:

5

6

7

8

Matrix 1:

1 2

3 4

Matrix 2:

5 6

7 8

Sum of two matrices:

6 8

10 12

2. Write a C program to read n number of values in an array and display them in reverse order.

CODE:

```
#include <stdio.h>
```

```
int main(){
```

```
    int arr[100];
```

```
    int n,i;
```

```
    printf("Enter the size of array:");
```

```
    scanf("%d",&n);
```

```
    printf("Enter %d elements: \n",n);
```

```
    for(i=0; i<n; i++){
```

```
        scanf("%d",&arr[i]);
```

```
    }
```

```
printf("Array elements in Reverse order:\n");  
for(i=n-1; i>=0; i--){  
    printf("%d ",arr[i]);  
}  
  
return 0;  
}
```

OUTPUT:

Enter the size of array:5

Enter 5 elements:

12

34

78

65

98

Array elements in Reverse order:

98 65 78 34 12

3.Count the total number of duplicate elements in an array.

CODE:

```
#include <stdio.h>
```

```
int main(){
```

```
int arr[100], n, i, j, dup_count = 0;

printf("Enter the size of array:");
scanf("%d",&n);

printf("Enter %d elements: \n",n);
for(i=0; i<n; i++){
    scanf("%d",&arr[i]);
}

for(i = 0; i < n; i++) {
    for(j = 0; j < i; j++) {
        if(arr[i] == arr[j]) {
            dup_count++;
            break;
        }
    }
}

printf("Total duplicate elements in array: %d\n",dup_count);

return 0;
}
```

OUTPUT:

Enter the size of array:5

Enter 5 elements:

11

12

11

43

12

Total duplicate elements in array: 2

4. Print unique elements in an array.

CODE:

```
#include <stdio.h>
```

```
int main(){
```

```
    int arr[100], n, i, j, count;
```

```
    printf("Enter the size of array:");
```

```
    scanf("%d",&n);
```

```
    printf("Enter %d elements: \n",n);
```

```
    for(i=0; i<n; i++){
```

```
        scanf("%d",&arr[i]);
```

```
    }
```

```
    printf("Unique elements in the array: ");
```

```
    for(i=0; i<n; i++){
```

```
        count=0;
```

```
        for(j=0; j<n; j++){
            if(arr[j] == arr[i]){
                count++;
            }
        }
        if(count == 1){
            printf("%d ", arr[i]);
        }
    }

    return 0;
}
```

OUTPUT:

Enter the size of array:8

Enter 8 elements:

12

34

56

12

11

12

10

68

Unique elements in the array: 34 56 11 10 68

5.Insert in sorted array.

CODE:

```
#include <stdio.h>

int main(){
    int arr[100],n,i,val;

    printf("Enter the size of sorted array:");
    scanf("%d",&n);

    printf("Enter %d elements: \n",n);
    for(i=0; i<n; i++){
        scanf("%d",&arr[i]);
    }

    printf("Enter the value:");
    scanf("%d",&val);

    for(i=n-1; i>=0 && arr[i]>val; i--){
        arr[i+1]=arr[i];
    }

    arr[i+1] = val;
    n++;
}
```

```
printf("New array: \n",n);  
for(i=0; i<n; i++){  
    printf("%d ",arr[i]);  
}  
  
return 0;  
}
```

OUTPUT:

Enter the size of sorted array:5

Enter 5 elements:

12

45

56

67

89

Enter the value:34

New array:

12 34 45 56 67 89

6.Insert in sorted array.

CODE:

```
#include <stdio.h>  
  
int main() {  
    int arr[100], n, i, pos, value, temp;
```

```
printf("Enter the size of array:");  
scanf("%d",&n);
```

```
printf("Enter the elements:");  
for (i=0; i<n; i++){  
    scanf("%d",&arr[i]);  
}
```

```
printf("Enter the position to insert:");  
scanf("%d",&pos);
```

```
printf("Enter the value:");  
scanf("%d",&value);
```

```
i = n - 1;  
while( i >= pos-1){  
    arr[i+1] = arr[i];  
    i--;  
}  
arr[pos-1] = value;  
n = n+1;
```

```
printf("New array:\n");  
for (i=0; i<n; i++){
```

```
        printf("%d ",arr[i]);  
    }  
  
    return 0;  
}
```

OUTPUT:

Enter the size of array:5

Enter the elements:12

34

56

78

90

Enter the position to insert:3

Enter the value:64

New array:

12 34 64 56 78 90

7.Matrix multiplication.

CODE:

```
#include <stdio.h>  
  
int main() {  
    int r1, c1, r2, c2, i, j, k;  
  
    printf("Enter rows of first matrix: ");
```

```
scanf("%d", &r1);
```

```
printf("Enter columns of first matrix: ");
```

```
scanf("%d", &c1);
```

```
printf("Enter rows second matrix: ");
```

```
scanf("%d", &r2);
```

```
printf("Enter columns of first matrix: ");
```

```
scanf("%d", &c2);
```

```
if (c1 != r2) {
```

```
    printf("Matrix multiplication is not possible.\n");
```

```
    return 0;
```

```
}
```

```
int A[r1][c1], B[r2][c2], C[r1][c2];
```

```
printf("Enter elements of first matrix:\n");
```

```
for (i = 0; i < r1; i++)
```

```
    for (j = 0; j < c1; j++)
```

```
        scanf("%d", &A[i][j]);
```

```
printf("Enter elements of second matrix:\n");
```

```
for (i = 0; i < r2; i++)
```

```
    for (j = 0; j < c2; j++)
```

```
        scanf("%d", &B[i][j]);
    for (i = 0; i < r1; i++)
        for (j = 0; j < c2; j++)
            C[i][j] = 0;

    for (i = 0; i < r1; i++) {
        for (j = 0; j < c2; j++) {
            for (k = 0; k < c1; k++) {
                C[i][j] += A[i][k] * B[k][j];
            }
        }
    }
    printf("Resultant Matrix:\n");
    for (i = 0; i < r1; i++) {
        for (j = 0; j < c2; j++) {
            printf("%d ", C[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

Enter rows of first matrix: 2

Enter columns of first matrix: 2

Enter rows second matrix: 2

Enter columns of first matrix: 2

Enter elements of first matrix:

1

2

3

4

Enter elements of second matrix:

5

6

7

8

Resultant Matrix:

19 22

43 50

8.Matrix transpose.

CODE:

```
#include <stdio.h>
```

```
int main(){
```

```
    int arr[10][10], rows, cols,i, j;
```

```
    printf("Enter size of row:");
```

```
    scanf("%d",&rows);
```

```
printf("Enter size of column:");
```

```
scanf("%d",&cols);
```

```
printf("Enter elements of the matrix:\n");
```

```
for(i=0; i<rows; i++){
```

```
    for(j=0; j<cols; j++){
```

```
        scanf("%d",&arr[i][j]);
```

```
    }
```

```
}
```

```
printf("\n");
```

```
printf("Matrix:\n");
```

```
for(i=0; i<rows; i++){
```

```
    for(j=0; j<cols; j++){
```

```
        printf("%d ",arr[i][j]);
```

```
    }
```

```
    printf("\n");
```

```
}
```

```
printf("Transpose of the matrix:\n");
```

```
for(i=0; i<cols; i++){
```

```
    for(j=0; j<rows; j++){
```

```
        printf(" %d ",arr[j][i]);
```

```
    }
```

```
    printf("\n");
```



```
}  
  
    return 0;  
}
```

OUTPUT:

Enter size of row:2

Enter size of column:3

Enter elements of the matrix:

12

34

45

56

67

78

Matrix:

12 34 45

56 67 78

Transpose of the matrix:

12 56

34 67

45 78

9.Right diagonal sum.

CODE:

```
#include <stdio.h>

int main(){

    int arr[10][10], n, i, j, sum=0;


    printf("Enter size of matrix:");

    scanf("%d",&n);


    printf("Enter elements of the matrix:\n");

    for(i=0; i<n; i++){

        for(j=0; j<n; j++){

            scanf("%d",&arr[i][j]);

        }

    }


    printf("\n");

    printf("Matrix:\n");

    for(i=0; i<n; i++){

        for(j=0; j<n; j++){

            printf("%d ",arr[i][j]);

        }

        printf("\n");

    }


    for(i = 0; i < n; i++) {
```

```
        for(j = 0; j < n; j++) {  
            if(i + j == n - 1) {  
                sum += arr[i][j];  
            }  
        }  
    }  
}  
  
printf("\nRight diagonal sum = %d", sum);  
  
return 0;  
}
```

OUTPUT:

Enter size of matrix:3

Enter elements of the matrix:

10

11

12

23

45

76

34

14

15

Matrix:

10 11 12

23 45 76

34 14 15

Right diagonal sum = 91

10.Check for identity matrix.

CODE:

```
#include <stdio.h>

int main(){
    int arr[10][10], n, i, j, found=1;

    printf("Enter size of matrix:");
    scanf("%d",&n);

    printf("Enter elements of the matrix:\n");
    for(i=0; i<n; i++){
        for(j=0; j<n; j++){
            scanf("%d",&arr[i][j]);
        }
    }

    printf("\n");
    printf("Matrix:\n");
```

```
for(i=0; i<n; i++){  
    for(j=0; j<n; j++){  
        printf("%d ",arr[i][j]);  
    }  
    printf("\n");  
}
```

```
for(i=0; i<n; i++){  
    for(j=0; j<n; j++){  
        if(i == j){  
            if(arr[i][j] !=1)  
                found =0;  
        }  
        else{  
            if(arr[i][j] != 0){  
                found = 0;  
            }  
        }  
    }  
}  
  
if(found==1){  
    printf("Identity matrix");  
}  
else{
```

```
        printf("Not an identity matrix");  
    }  
    return 0;  
}
```

OUTPUT:

Enter size of matrix:3

Enter elements of the matrix:

1

0

0

0

1

0

0

0

1

Matrix:

1 0 0

0 1 0

0 0 1

Identity matrix

11.To find out number of vowels, consonants, digits, and white spaces.

CODE:

```
#include <stdio.h>

int main(){
    char str[100];
    int i;
    int digits =0, vowels=0, con=0, spaces=0;
    printf("Enter the string: ");
    fgets(str, sizeof(str), stdin);

    for(i=0; str[i] != '\0'; i++){
        if(str[i] >='0' && str[i] <='9'){
            digits++;
        }
        else if(str[i] == 'a' ||str[i] == 'e' ||str[i] == 'i' ||str[i] == 'o' ||str[i] == 'u' ||str[i]
== 'A' ||str[i] == 'E' ||str[i] == 'I' ||
        str[i] == 'O' ||str[i] == 'U'){
            vowels++;
        }
        else if(str[i] == ' '){
            spaces++;
        }
        else if((str[i] >= 'A' && str[i] <= 'Z') ||
            (str[i] >= 'a' && str[i] <= 'z')) {
            con++;
        }
    }
}
```

```

    }

}

printf("Digits: %d\nVowels: %d\nConsonants: %d\nWhite spaces:
%d",digits, vowels, con, spaces);

return 0;

}

```

OUTPUT:

Enter the string: I am very much interested in Problem solving 5028

Digits: 4

Vowels: 13

Consonants: 24

White spaces: 8

12.Frequency of character in a string.

CODE:

```

#include <stdio.h>

int main(){

    char str[100];

    int i, freq[26] = {0};


    printf("Enter the string: ");

    fgets(str,sizeof(str), stdin);


    for(int i =0; str[i] !='\0'; i++){

        if(str[i] >= 'A' && str[i] <= 'Z' ){

```



```

        str[i] = str[i] + 32;
    }
}
for(int i =0; str[i] !='\0'; i++){
    freq[str[i] - 'a']++;
}
for (i = 0; i < 26; i++) {
    if (freq[i] > 0){
        printf("%c = %d\n", i + 'a', freq[i]);
    }
}
return 0;
}

```

OUTPUT:

Enter the string: programming

a = 1

g = 2

i = 1

m = 2

n = 1

o = 1

p = 1

r = 2

13.Count words in a string.

CODE:

```
#include <stdio.h>

int main(){
    char sen[100];
    int i, words = 1;
    printf("Enter the sentence:");
    fgets(sen, sizeof(sen), stdin);
    if(sen[0] == '\0' || sen[0] == '\n'){
        words = 0;
    }
    for( i = 0; sen[i] != '\0'; i++){
        if(sen[i] == ' ' && sen[i+1] != ' ' && sen[i+1] != '\0'){
            words++;
        }
    }
    printf("Number of words in the given sentence:%d", words);
    return 0;
}
```

OUTPUT:

Enter the sentence:I am very much interested in Problem solving

Number of words in the given sentence:8

14.Sort string array.

CODE:

```
#include <stdio.h>

#include <string.h>

int main() {

    char str[10][50], temp[50];

    int n, i, j;

    printf("Enter the number of strings: ");

    scanf("%d", &n);

    getchar();

    printf("Enter the strings:\n");

    for(i = 0; i < n; i++) {

        fgets(str[i], sizeof(str[i]), stdin);

        str[i][strcspn(str[i], "\n")] = '\0';

    }

    for(i = 0; i < n - 1; i++) {

        for(j = i + 1; j < n; j++) {

            if(strcmp(str[i], str[j]) > 0) {

                strcpy(temp, str[i]);

                strcpy(str[i], str[j]);

                strcpy(str[j], temp);

            }

        }

    }

}
```

```

        }
    }
}

printf("\nSorted strings:\n");
for(i = 0; i < n; i++) {
    printf("%s\n", str[i]);
}

return 0;
}

```

OUTPUT:

Enter the number of strings: 3

Enter the strings:

Mango

Grapes

Orange

Sorted strings:

Grapes

Mango

Orange

15.Toggle case of a sentence.

CODE:

```
#include <stdio.h>

int main(){

    char str[100];

    printf("Enter string:");

    fgets(str,sizeof(str),stdin);


    for(int i = 0; str[i] != '\0'; i++){

        if(str[i] >= 'A' && str[i] <= 'Z'){

            str[i] = str[i] + 32;

        }

        else if(str[i] >= 'a' && str[i] <= 'z'){

            str[i] = str[i] - 32;

        }

    }

    printf("Case changed string:%s",str);

    return 0;

}
```

OUTPUT:

Enter string:HELlo woRLd

Case changed string:helLO WORld

